

LEAKLISTEN · REQUIREMENTS

LeakListen requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/03-requirements.md @ 35e4499		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

LeakListen requirements

These are first-pass requirements for one logger. Targets are proposals for review; they come from desk research, not yet from a partner utility, and must be revised after co-design (see the problem statement, LKL-PRB-001). Status is judged against the concept in LKL-PRC-001 and is an estimate until checked at TRL 3 or later.

Table 1. Requirements and status at TRL 2.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Hear a leak on metallic mains	Flag a 5 L/min (1.3 US gal/min) leak at 3 bar within 100 m of the logger on cast or ductile iron DN100 to DN200	Literature attenuation estimate at TRL 3; later a test rig or partner site with a controlled leak	Unverified
R2	Hear a leak on plastic mains	Same leak within 30 m on PVC or PE DN100 to DN200	As R1	At risk: plastic pipes attenuate leak noise strongly
R3	Sensor bandwidth	5 Hz to 2 kHz, within ± 3 dB after correction	Preamplifier calculation; later a shaker comparison with a reference accelerometer	Met on paper (design choice)
R4	Sensor self-noise	Equivalent input noise 1 $\mu\text{g}/\sqrt{\text{Hz}}$ or less from 100 Hz to 1 kHz	Noise calculation of the piezo and charge amplifier at TRL 3	Unverified
R5	Night listening	12 windows of 20 s between 02:00 and 04:00 local time, clock drift under 1 min per month	Firmware sketch and RTC datasheet	Met on paper
R6	Battery life	5 years or more on one primary cell at the R5 schedule and one uplink a night	Power budget calculation	Met on paper: about 10 years (estimate)
R7	Radio link	90 % or more of nightly summaries delivered to a gateway within 1 km from a chamber with a cast-iron cover	Link budget at TRL 3; later a field check	At risk: cover loss is unknown and may block the link
R8	Data and privacy	Only band levels and spectra leave the logger; raw vibration samples are deleted on the device after processing	Design review of the firmware sketch	Met by design
R9	Attachment	Magnet holds 100 N or more on a steel or iron spindle cap, with no tools or pipe work	Magnet datasheet and a pull test later	Met for ferrous fittings; not met for brass, bronze or plastic fittings
R10	Install from the surface	Placed and recovered in 10 min or less without entering the chamber, for chambers up to 1.5 m deep	Walk-through with the partner utility	Met on paper; not yet shown in the field
R11	Survive the chamber	IP68: 1 m submersion for 7 days; operate from -20 to +50 °C	Design review; later a submersion test	Met on paper (sealed tube, IP68 connector)
R12	Size and mass	Logger 70 mm diameter or less and 300 mm long or less; total mass 1.0 kg or less	Massing model	Met: 63 x 240 mm, about 0.7 kg (estimate)
R13	Leak flag quality	1 false alarm or fewer per 20 loggers per month after a 14-night baseline	Needs field data from a partner network	Unverified
R14	Open and interoperable		Design review	Met by design

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
		Standard LoRaWAN 1.0.x uplink, documented payload, works with any network server and TwinKit		
R15	Cost	Parts cost \$130 or less per logger	Priced BOM	Met: about \$108 (indicative)

■ Requirements not met or at risk

- **R2 at risk.** Leak noise attenuates quickly on plastic pipes ([Hunaidi and Chu, 1999](#)), so 30 m may not be reachable with a low-cost sensor.
- **R7 at risk.** A cast-iron cover may block the LoRa link. Options (proposed, awaiting Amish): a composite cover, a through-cover antenna, or a lower spreading factor gateway placed nearby.
- **R9 not met** for non-ferrous fittings. A strap-on adapter would be needed.
- **R1, R4 and R13 unverified.** They depend on sensor noise, pipe attenuation and site background noise, which need calculation at TRL 3 and field data later.

■ Assumptions

- Network pressure at night 2 to 6 bar; leaks at lower pressure are quieter.
- Valve spindle caps and hydrant bodies are cast iron or steel on most networks the lab would target first.
- A LoRaWAN gateway (TwinKit, a partner's or a public network) is within about 1 km in an urban area.
- Primary cell: lithium thionyl chloride C size, 3.6 V, about 7.7 Ah nominal, derated to 70 % usable for cold and pulse loads.

■ Safety requirements

Safety: S1: no confined space entry is needed to place, service or recover a logger. S2: the cell is a primary lithium cell with a fuse or PTC and reverse protection; no charging circuit is fitted. S3: the logger carries a label with the owner, a contact and a lithium cell warning. S4: nothing touches drinking water, and nothing is fixed to the pipe other than the magnet.